

Group 9 Assignment Report

An Evolutionary Algorithm for the Travelling Salesman Problem

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1 Introduction

The goal of this assignment is to design, implement and experimentally evaluate an evolutionary algorithm for a combinatorial optimisation problem. We chose the Travelling Salesman Problem, whose discrete, permutation-based search space and relevance to routing and logistics make it a natural and challenging testbed for evolutionary computation. This report presents our problem formulation, the evolutionary approach and operators we developed, and an experimental analysis of its solution quality and convergence behaviour.

2 The Travelling Salesman Problem

2.1 Mathematical Formulation

The Travelling Salesman Problem (TSP) is a classical NP-hard combinatorial optimisation problem: given a set of N cities and the distances between them, the task is to find the shortest closed tour that visits every city exactly once. Following the formulation of Potvin [1], we model the problem on a complete weighted graph $G = (V, E)$, where $V = \{1, \dots, N\}$ is the set of cities and d_{ij} is the distance between cities i and j . The decision variables are $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if and only if the tour travels directly from city i to city j . The objective is to minimise the total travelled distance,

$$\text{Minimise} \quad \sum_{i=1}^N \sum_{j=1}^N d_{ij} x_{ij}, \quad (1)$$

subject to the degree constraints

$$\sum_{j=1}^N x_{ij} = 1, \quad i = 1, \dots, N, \quad \sum_{i=1}^N x_{ij} = 1, \quad j = 1, \dots, N, \quad (2)$$

which require every city to have exactly one incoming and one outgoing edge, and the subtour-elimination constraints

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1, \quad \forall S \subset V, \quad 2 \leq |S| \leq N - 1, \quad (3)$$

which forbid partial cycles over any proper subset S of the cities.

Because an evolutionary algorithm operates on orderings rather than on binary edge variables, it is convenient to state the equivalent tour-as-permutation formulation. A candidate solution is a permutation $\pi = (\pi_1, \dots, \pi_N)$ of the cities, and its cost is the length of the cycle that visits them in that order,

$$C(\pi) = \sum_{k=1}^{N-1} d_{\pi_k, \pi_{k+1}} + d_{\pi_N, \pi_1}. \quad (4)$$

The evolutionary algorithm therefore seeks a permutation minimising Equation (4).

2.2 Related Evolutionary Approaches

Liu [2] builds a genetic algorithm around a crossover operator called *Edge Swapping* (ES), which keeps the edges shared by the two parent tours and swaps in the cycles where they disagree; seeding the population with greedy tours refined by 2-opt, it reaches best-known solutions on large benchmarks. Liao et al. [3] propose GPSO, a genetic-based particle swarm optimisation in which each particle is a real-valued vector decoded into a tour by ranking its components, initialised geometrically by sorting cities by their angle around the geographical centre, which produces a non-crossing route. For the path representation, where a chromosome is the visiting order and classical one-point or uniform crossovers cannot be used as they duplicate cities, Hussain et al. [4] compare partially mapped (PMX), order (OX) and cycle crossover, where a segment of one parent is copied into the child and the remaining cities are filled in their order of appearance in the other parent; no operator dominates on TSPLIB.

2.3 Motivation of the Developed Approach

The evolutionary approach we develop follows the genetic algorithm presented in Section 9.2 of Mitchell [6], adapted to the permutation representation of the TSP; its pseudo-code is given in Section 3. Within this template, the design choices left open concern the initialization of the population and the crossover operator, which we align with the literature reviewed in Section 2.2. We test four initializations: random, greedy nearest-neighbour tours refined by 2-opt [2], geometric non-crossing tours [3], and their combination. For recombination we adopt the order crossover (OX) [4]. The remaining operators are left unchanged, so that the effect of the two adapted components can be isolated in Section 4.

3 Algorithm

Each individual is represented as a permutation of all cities, where the order of the cities defines the tour and the final city is connected back to the first. The initial population consists of p tours generated by the selected initialization method. The cost of an individual is given by Equation (4). Since the TSP is a minimisation problem, this cost is converted into a maximisation-based fitness function defined as

$$F(\pi) = \frac{1}{C(\pi)}. \quad (5)$$

Consequently, shorter tours receive higher fitness values and are favoured during selection. In the implementation, tour costs are used directly where possible, since minimizing $C(\pi)$ is equivalent to maximizing $F(\pi)$.

In each generation, approximately a fraction $(1 - r)$ of the population is selected deterministically by retaining the individuals with the lowest tour costs. These individuals are copied to the new population before mutation. The remaining fraction is filled with offspring generated from selected parent pairs.

Parent selection for crossover is performed using tournament selection. For each parent, k individuals are sampled randomly from the current population, and the individual with the lowest tour cost is selected. The process is repeated to obtain the second parent. Tournament selection provides selection pressure towards shorter tours while still allowing different individuals to participate in reproduction.

Order Crossover (OX) is used to generate offspring because it preserves valid permutations. A segment from one parent is copied to the offspring, while the remaining cities are inserted

according to their order in the second parent, skipping already used cities. Two offspring are generated for each parent pair by swapping the parent roles.

After crossover, mutation is applied to a fraction m of the new population. The individuals to be mutated are selected uniformly at random. Depending on the experiment, either swap mutation or inversion mutation is used. Swap mutation exchanges two randomly selected positions in a tour, whereas inversion mutation reverses the order of the cities within a randomly selected segment. Both mutation operators preserve a valid permutation.

Once mutation has been applied, the new population replaces the previous one and all tour costs are evaluated again. The best tour found so far is stored throughout the run. This means that even if a previously good individual is later modified by mutation or disappears from the population, the best solution encountered during the search is not lost.

The process is repeated for G generations. After termination, the best tour encountered during the complete run is returned.

Algorithm 1 Genetic Algorithm for the Travelling Salesman Problem

Require: cities D , population size p , replacement rate r , mutation rate m , number of generations G , tournament size k **Ensure:** best tour found $P \leftarrow$ generate p initial toursevaluate the cost of each individual in P $best \leftarrow$ individual in P with minimum cost**for** $g = 1$ to G **do** create new generation $P_s \leftarrow \emptyset$ **Selection:** $q \leftarrow \text{round}(rp/2)$ $s \leftarrow p - 2q$ add the best s individuals from P to P_s **Crossover:** **for** $i = 1$ to q **do** select h_1 from P using tournament selection of size k select h_2 from P using tournament selection of size k $o_1 \leftarrow \text{ORDERCROSSOVER}(h_1, h_2)$ $o_2 \leftarrow \text{ORDERCROSSOVER}(h_2, h_1)$ add o_1 and o_2 to P_s **end for** **Mutation:** uniformly select mp individuals from P_s **for** each selected individual π **do** apply the selected mutation operator to π **end for** **Update:** $P \leftarrow P_s$ **Evaluation:** evaluate the cost of each individual in P **if** the best individual in P has lower cost than $best$ **then** update $best$ **end if** **end for** **return** $best$

4 Experimental part

4.1 Hardware and Software

All experiments were run on two MacBook Pro laptops with Apple Silicon (M-series) processors. The algorithm is implemented in Python 3.12 using only the standard library; `matplotlib` is used solely to produce the figures. Every trial runs as a single process, and all randomness is drawn from a seeded generator, so a fixed seed reproduces identical results on any machine.

4.2 Experimental Protocol

Every configuration is repeated in several independent runs seeded from a fixed master seed. The study proceeds in three stages. First, the random, nearest-neighbour/2-opt, geometric (angle-based) and combined initialisations are compared on burma14, berlin52, kroA100 and kroA200. The nearest-neighbour and combined ones already start at the optimum on burma14 and berlin52 (mean gap 0.00%) and within about 2% of it on kroA100 and kroA200, and the GA improves them by less than 0.3 percentage points, whereas the random and angle-based ones improve substantially. Second, each hyperparameter is swept on the same four instances, one at a time: population size $p \in \{50, 100, 120, 150\}$, generations $G \in \{100, 300, 500, 800\}$, mutation rate $m \in \{0.05, 0.1, 0.2, 0.4\}$ and replacement rate $r \in \{0.3, 0.5, 0.7, 0.9\}$; swap and inversion mutation are also compared on berlin52, and inversion is retained. Each sweep is repeated for all four initialisations and varies one parameter around the default configuration ($p = 100$, $r = 0.7$, $m = 0.1$, $G = 500$). Third, the problem-size scaling experiment repeats this configuration on the six instances, using the random initialisation.

4.3 Datasets

The evaluation uses six symmetric TSPLIB instances [5] of increasing size: burma14 (14 cities), berlin52 (52), kroA100 (100), kroA200 (200), pcb442 (442) and pr1002 (1002). They are committed to the repository and loaded with their original TSPLIB edge weights (GEO for burma14, EUC_2D for the others). Performance is measured as the optimality gap $(C_{\text{best}} - C_{\text{opt}})/C_{\text{opt}} \cdot 100$, using the published optima 3323, 7542, 21282, 29368, 50778 and 259045. The main study uses berlin52; the scaling experiment covers all six instances.

5 Results and Analysis

6 Conclusions

References

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